

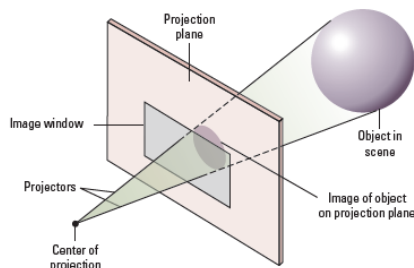
The Scope of Computer Graphics

- **Input is data, Output is graphic**
- **2D Graphics:** Deals with manipulating **two-dimensional images**
- **3D Graphics:** Deals with producing and displaying images of **three-dimensional virtual scenes**.

1. Modeling

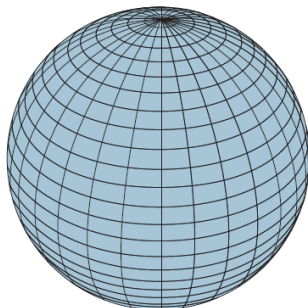
- **Scene:** Objects(物件)、Light、Camera

2. Randerimg(渲染)

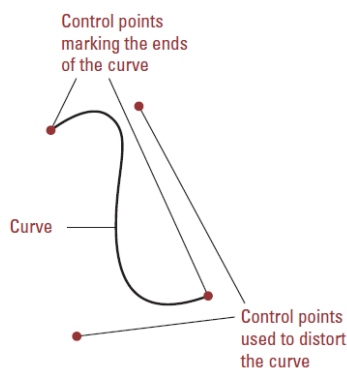


Modeling Objects

- **Shape**
 - **Polygon Mesh(多邊形網格)**
 - 3D Object is consist of many **planar patches**.



- **Curve: Bézier Curves(2D)**
 - **Bézier Surface(3D):** 2D → 3D



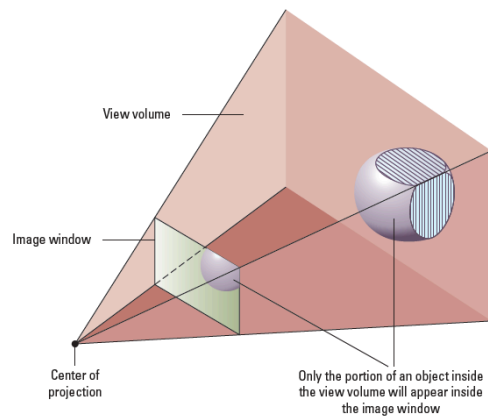
- **Procedural Models(程序化建模):** Use **algorithms** to produce. e.g. mountain, plant, particle...
- **Surface:** Be represented by a **texture map**.

Randerimg

- **Reflection:** Light rays bounce off surface.
 - **Specular Light(鏡面反射光/高光):** Color of the light source
 - **Diffuse Light(漫反射):** Color of the object
 - **Ambient Light(環境光):** Shadow
- **Refraction:** Light rays penetrate surface
- **Rendering Pipeline**
 1. **Clipping:** Restricts attention to objects within **view volume(攝影範圍)**.
 2. **Scan Conversation:**
 - **Rasterization(光柵化):** vector graph turns into pixel graph

3. **Shading**: Determines **appearance** of points associated with pixels.

- **Flat Shading(平面著色)**: Creates faceted appearance.
- **Gouraud and Phong Shading(平滑著色)**: Creates smooth, rounded appearance.
- **Bump Mapping(凹凸貼圖)**: Creates bumpy, rounded appearance.



Local & Global Illumination

- **Local Illumination**: Does **not account for light interactions among objects**.
- **Global Illumination**: Accounts for light interactions among object.
 - **Ray Tracing**
 - **Recursive process**: Eye → Object → Light Source
 - **Radiosity(輻射度演算法)**: Divide object surface into many parts and assign the light to each every part.

Animation

- **Storyboard(分鏡腳本)**: A sequence of sketches summarizing the entire animation.
- **Frame(影格)**: One of many images used to create animation.
- **Key Frames(關鍵影格)**: Frames capturing the scene at specified points in time.
- **In-betweening(補間)**: Producing frames to fill the gaps between key frames.

Simulation Motion

- **Dynamics**: Applies laws of physics to determine position of objects.
- **Kinematics**: Applies characteristics of joints and appendages to determine position of objects.
 - **Avars**
 - **Motion Capture**